Like terms and their collection

Only terms with the same letters can be added — and that one rule tidies every expression.

LESSON 69–70 | 2 × 40 MIN | MATEMATIKA 7, §32 | S8 2.2

LESSON CLOCK

15' | 25' | 25' | 20' | 5'

Which terms are alike	15 min
Collecting	25 min
After brackets	25 min
In problems	20 min
Homework	5 min

LEARNING OBJECTIVES

- Recognise like terms.
- Collect like terms, including with negative coefficients.
- Simplify an expression after removing brackets.
- Use collection to shorten a perimeter or a cost calculation.

TEXTBOOK

National	pp. 193–198
Cambridge	Stage 8, pp. 16–20
Workbook	Exercise 2.2

01 Explanation

Which terms are alike

Two terms are **like** if their letter parts are identical — the same letters with the same exponents. The coefficients may be anything.

PAIR	LIKE?	WHY
$3x, -7x$	yes	same letter part x
$4a^2b, a^2b$	yes	same letter part
$4a^2b, 4ab^2$	no	different exponents
$5x, 5y$	no	different letters
$7, -2$	yes	both constants

THINK OF THE LETTER PART AS A UNIT

Three apples plus five apples is eight apples; three apples plus five pears is neither. That is the whole rule, and it explains why unlike terms simply stay as they are.

Collecting

$$ka + la = (k + l)a$$

Add the coefficients and keep the letter part.

EXPRESSION	COLLECTED
$3x + 5x$	$8x$
$7a - 10a$	$-3a$
$4x + 3y - x + 5y$	$3x + 8y$
$2a^2 + 3a - a^2 + a$	$a^2 + 4a$

TAKE THE SIGN WITH THE TERM

In $4x + 3y - x + 5y$ the third term is $-x$, so the x terms give $4 - 1 = 3$. Copying a term without its sign is the commonest error here.

After brackets

Remove the brackets first, then collect.

EXPRESSION	BRACKETS REMOVED	COLLECTED
$2(x + 3) + 3(x - 1)$	$2x + 6 + 3x - 3$	$5x + 3$
$5(a - 2) - 2(a - 5)$	$5a - 10 - 2a + 10$	$3a$
$x - (3x - 4)$	$x - 3x + 4$	$-2x + 4$

TWO PASSES, NEVER ONE

Removing brackets and collecting at the same time is where the signs go wrong. Write the expanded line in full before touching the collection.

In problems

SITUATION	EXPRESSION	SIMPLIFIED
perimeter of a rectangle a by b	$a + b + a + b$	$2a + 2b$
perimeter of a triangle $x, x, 2x - 1$	$x + x + 2x - 1$	$4x - 1$
3 pens at p and 4 books at b, twice over	$2(3p + 4b)$	$6p + 8b$

SIMPLIFYING IS WHAT MAKES A FORMULA USABLE

$2a + 2b$ can be evaluated in two operations; $a + b + a + b$ needs three. Over a whole calculation the saving is large, and the shorter form is also easier to read.

02 Worked examples

EXAMPLE 1 **Collect $4x + 3y - x + 5y$.**

1. x terms: $4x - x = 3x$.

   The sign belongs to the term.

2. y terms: $3y + 5y = 8y$.

3. They cannot be combined further.

4. $3x + 8y$

ANSWER

$3x + 8y$

EXAMPLE 2 **Simplify $5(a - 2) - 2(a - 5)$.**

1. $5a - 10$ and $-2a + 10$.
   Both signs change in the second.
2. $5a - 10 - 2a + 10$
3. a terms: $3a$; constants: 0.
4. $= 3a$

ANSWER

$3a$

EXAMPLE 3 **Find and simplify the perimeter of a triangle with sides x, x and $2x - 1$.**

1. $P = x + x + (2x - 1)$
2. $= x + x + 2x - 1$
3. Collect: $4x - 1$.
4. Check at $x = 3$: $3 + 3 + 5 = 11 = 4(3) - 1$ ✓

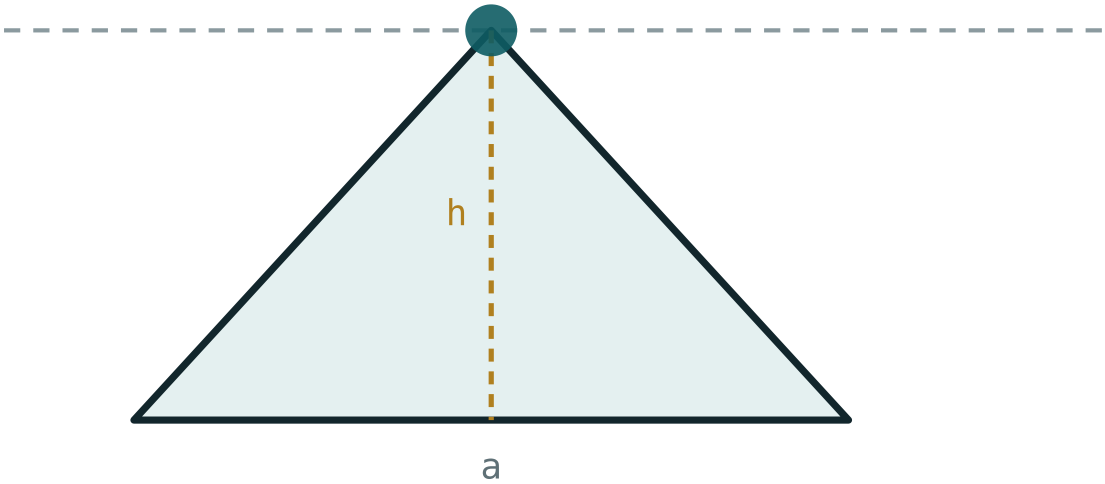
ANSWER

$4x - 1$

03 Interactive model

Ask for the perimeter of a rectangle twice — once unsimplified and once collected — and have the class evaluate both at a = 7, b = 4; the answers agree and one is far quicker.

Collecting like terms



base a **220** | height h **120** | area ½ah **13200** | perimeter **545.6**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Like terms	O‘xshash hadlar	Подобные члены
To collect	Ixchamlash	Приведение подобных
Unlike terms	O‘xshash bo‘lmagan hadlar	Неподобные члены
Coefficient	Koeffitsiyent	Коэффициент
Letter part	Harfiy qism	Буквенная часть
To simplify	Soddalashtirish	Упростить
Perimeter	Perimetr	Периметр
Cost	Narx	Стоимость

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check

Check yourself

Are $4a^2b$ and $4ab^2$ like terms?

- yes
- no
- sometimes
- only if $a = b$

$3x + 5x$ equals:

- $8x$
- $8x^2$
- $15x$
- 8

$7a - 10a$ equals:

- $3a$
- $-3a$
- $17a$
- $-17a$

$4x + 3y - x + 5y$ equals:

- $3x + 8y$
- $5x + 8y$
- $3x + 2y$
- $11xy$

$5(a - 2) - 2(a - 5)$ equals:

- $3a - 20$
- $3a$
- $7a - 20$
- $3a + 20$

Can $3x$ and $5y$ be collected?

- yes
- no
- as $8xy$
- as $15xy$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up 7 PROBLEMS

1. $3x + 5x$

   ANSWER $8x$

2. $7a - 10a$

   ANSWER $-3a$

3. $2b + b$

   ANSWER $3b$

4. $5x - 5x$

   ANSWER 0

5. $4a^2 + 3a^2$

   ANSWER $7a^2$

6. Can $3x$ and $5y$ be collected?

   ANSWER No

7. Are 7 and -2 like terms?

   ANSWER Yes

Medium · the standard question

7 PROBLEMS

1. $4x + 3y - x + 5y$

   ANSWER $3x + 8y$

2. $2a^2 + 3a - a^2 + a$

   ANSWER $a^2 + 4a$

3. $2(x + 3) + 3(x - 1)$

   ANSWER $5x + 3$

4. $5(a - 2) - 2(a - 5)$

   ANSWER $3a$

5. $x - (3x - 4)$

   ANSWER $4 - 2x$

6. Perimeter of a rectangle a by b

   ANSWER $2a + 2b$

7. Perimeter of a triangle $x, x, 2x - 1$

   ANSWER $4x - 1$

Hard · stretch and reason

7 PROBLEMS

1. $3(2a - b) - 2(3a - 2b)$

   ANSWER b

2. $4x^2 - 3x + 1 - (x^2 - 3x + 5)$

   ANSWER $3x^2 - 4$

3. $a - [2a - (3a - b)]$

   ANSWER $2a - b$

4. $2(x + y) - 3(x - y) + x$

   ANSWER $5y$

5. Perimeter of a square of side $3x - 2$

   ANSWER $12x - 8$

6. A rectangle $x + 3$ by $x - 3$: its perimeter

   ANSWER $4x$

7. Simplify $5ab - 2ba$

   ANSWER $3ab$

07 Homework

Homework — 5 tasks

Write the expanded line in full before collecting anything.

1. Collect $6a + 2b - 4a + 7b$.
2. Simplify $3(x + 4) - 2(x - 1)$.
3. Simplify $y - (4y - 7)$.
4. Find and simplify the perimeter of a rectangle $2x + 1$ by $x - 2$.
5. Simplify $5m^2 + 2m - 3m^2 - 6m$.